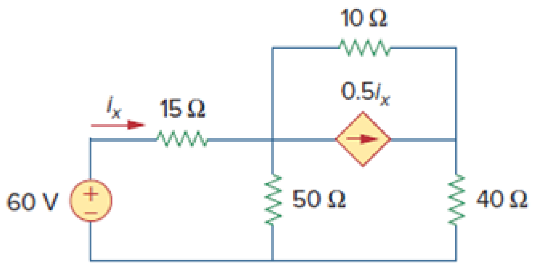
Problem

Use source transformation to find *ix* in the circuit of Fig. 4.100.

Figure. 4.100:



Step-by-step solution

Step 1 of 7

As shown in Fig-(a), we transform the dependent current source to a voltage source,

Step 2 of 7

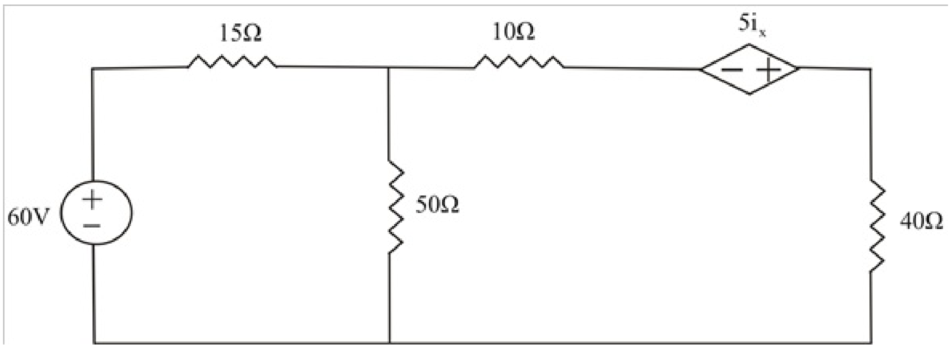
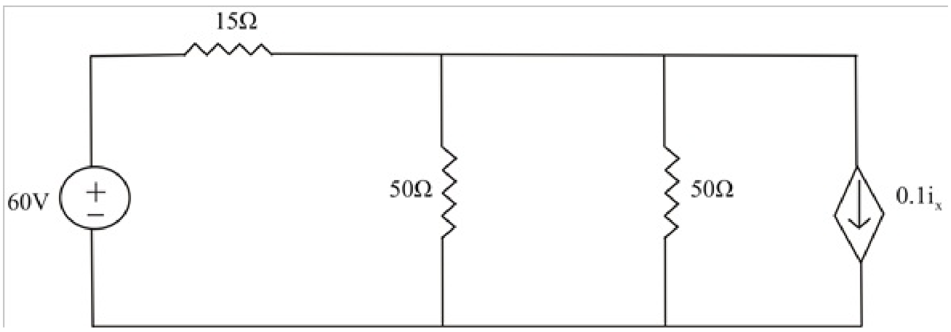


Fig-(a)

Step 3 of 7

As shown in Fig-(b), we transform the dependent voltage source to a current source.



Step 4 of 7

Fig-(b)

Step 5 of 7

$$50\Omega \| 50\Omega = \frac{50\times 50}{50+50}$$

$$50\Omega \| 50\Omega = \frac{2500}{100}$$

$$50\Omega \| 50\Omega = 25\Omega$$

Step 6 of 7

As shown in Fig-(c), we transform the dependent current source to a voltage source,

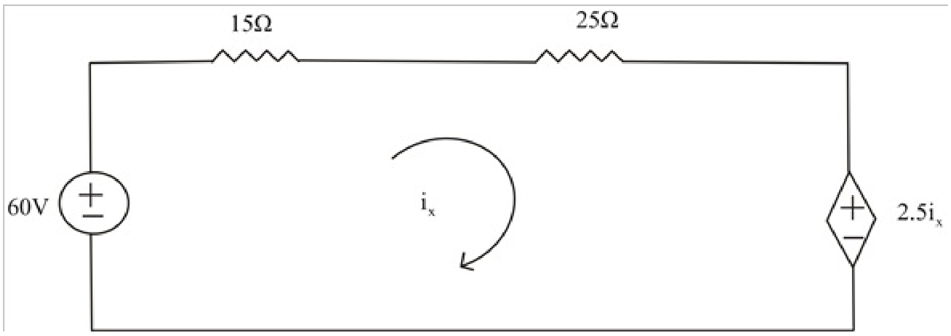


Fig-(c)

Step 7 of 7

Applying KVL in Fig-(c),

$$-60+40i_x-2.5i_x=0$$

$$-60=-37.5i_x$$

$$\therefore \boxed{i_x=1.6\,\text{A}}\ .$$